

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

5ARE0 — Data Analysis & Learning Methods | Assignment 1 | Group 5

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1 Introduction

The CRISP-DM (CRoss-Industry Standard Process for Data Mining) framework described by [1] was the underlying reference for this project, associated with the design directives and studying material from this course.

The group adopted an experimental approach to investigate the data and establish frequencies of interest related to the type of road. The empirical hypothesis was that a so called 'bumpy' road would, by definition, present a higher energy in the band of interest due to vertical vibration, which is basically the rider's perceived shaking experience when cycling through a bumpy versus a smooth path.

2 Business / Objective Understanding

If a rider could close their eyes while cycling (which is not recommended by obvious reasons), this person's body would be able to clearly feel that they are passing either in a smooth or in a bumpy road. Perhaps one of the easiest ways to explain that in physics-related terms would be that when one is crossing a bumpy path, the dynamic vertical acceleration of one's bicycle is much more noticeable (and uncomfortable) if compared to a completely smooth terrain.

This definition gives us a strong hypothesis to test. If this very distinctive vibration feeling that the riders experiment can be derived from the sensors already present in a smartphone, for instance, that would be a clear and well-defined way to feed an autonomous classifier to ingest and analyse the sensor data, providing a real-time diagnostic whether a bike path is smooth or bumpy.

Naturally, this diagnostic is irrelevant on a user by user basis. As we already established, it's quite easy to feel the condition of the road you are cycling in. Accordingly, the value of such a classifier would be perceived for stakeholders responsible for providing adequate road conditions, or even more so, to equip the community with a tool to evaluate which parts of a given municipality need smoother cycling paths.

Considering the actual classifier, what is important to determine and establish in terms of scope is the ability of the model to separate between two distinct classes: *smooth* and *bumpy*. So, let's define that a single pothole in a smooth bicycle lane **should not** necessarily trigger a bumpy classification. As much as a pothole needs fixing to avoid accidents, this should fall in routine repairs and, as stated, the objective here is to raise awareness to bumpy lanes that need to be, at some point, converted to smooth ones.

Research objective. As a high level model design criteria, the targets were defined as: "*Smooth*" = 0 and "*Bumpy*" = 1. Since the main objective is to locate paths which are bumpy across cities and villages, what is most important for this model is to avoid classifying smooth roads as bumpy, in other words, we need a model focused on delivering high precision bumpy outcomes, supported by a low $\frac{\text{false positive rate}}{\text{specificity}}$, where a false positive is defined as: actual Smooth $\rightarrow$ predicted Bumpy.

- headline metric = Bumpy precision
- direct FP metric = Smooth specificity / false-positive rate
- anti-degenerate check = Bumpy recall + F0.5

It is important to notice that we should not optimise precision alone and then hide recall. A model that would never predict 'Bumpy' can obtain excellent precision, while being useless in practice.

Technology Statement: During the preparation of this work, we used OpenAI ChatGPT (GPT-5.6 Sol), Google Gemini (3.1 Pro) and Claude Sonnet 5 Medium in order to brainstorm assignment direction, support on coding and discuss specific technical concepts to support the team along the development. The following parts of the assignment were affected/generated by AI tool usage: [DATA PREPARATION / MODELING / EVALUATION]. After using this tool/service, [Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto] evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, [Su Acar, Vikram Chandrasekaran and Henrique da Silva Couto] take full responsibility for the content of their work.

3 Data Understanding

3.1 Acquisition Protocol

Acquisition protocol. *[Insert: Participants, anonymization, back-pocket placement, accelerometer/gravity/gyroscope configuration, number/duration/terrain of rides, and source organization.]*

Quality screening. *[Insert: Show measured sampling rates; document exclusion of participant #03 (62 Hz, Nyquist 31 Hz, incompatible with 40–46 Hz features). Describe what this does to participant generalizability.]*

Available independent units. *[Insert: State both number of windows and independent original recording/session groups; identify how Raw031–Raw041 map to two parent sessions.]*

3.2 Empirical Measurements - Road Frequency versus Leg Movement Artifacts

It was decided to conduct an experiment to evaluate whether there was a possible overlap with low frequency artifacts (such as leg movements) with the natural vibration caused by the gaps on the road.

Accordingly, an empirical experiment was carried out to empirically measure those frequencies.



Figure 1: Example of manually counting the rotations per minute based on the video recording from one of the participants.



Figure 2: Different types of bumpy terrains were considered. The measured terrains ranged from 100 ~ 500mm gap interval.

The approximate road-excitation frequency was estimated from the measured cycling velocity and the spatial spacing between terrain irregularities:

$$f_{\text{road}} \approx \frac{v}{\lambda}, \quad (1)$$

where v is the bicycle velocity and λ is the spatial terrain spacing. For the experiment, $\lambda = 100 \text{ mm} = 0.1 \text{ m}$. The bicycle velocity was calculated from the travelled distance d and the mean time of the three repetitions:

$$v = \frac{d}{t}. \quad (2)$$

Pedalling frequency was estimated by manually counting complete leg-motion cycles over a measured video duration:

$$f_{\text{pedal}} = \frac{N_{\text{cycles}}}{t}, \quad (3)$$

with cadence expressed in revolutions per minute as

$$\text{RPM} = 60f_{\text{pedal}}. \quad (4)$$

Table 1: Measured cycling speed and estimated road-excitation frequency for a terrain spacing of $\lambda = 100$ mm.

λ (mm)	Distance (m)	Intended speed	t_1 (s)	t_2 (s)	t_3 (s)	$\bar{t}$ (s)	Speed (m/s)	f_{road} (Hz)
100	10	Slow	3.63	4.06	4.03	3.91	2.56	25.60
100	10	Medium	2.50	2.45	2.60	2.52	3.97	39.74
100	10	Fast	1.95	1.85	2.06	1.95	5.12	51.19

As shown in table 1, the data was calculated for a gap interval $\lambda = 100\text{mm}$. As it can be seen in figure 2, we can easily extrapolate those results with $\lambda = 300\text{mm}$ and $\lambda = 500\text{mm}$.

We already know the higher end of the spectrum, with $\lambda = 100\text{mm}$ at maximum speeds, which yields a road frequency of $\approx 51\text{Hz}$. To estimate to the best of our abilities the other side of the spectrum, we need to consider the maximum gap measured ($\lambda = 500\text{mm}$), associated with the slowest speed in table 1 ($\approx 2.5\text{m/s}$) which will, in turn, result in a minimum road frequency of $\approx 5\text{Hz}$.

Table 2: Estimated pedalling cadence from manually counted leg-motion cycles.

Video	Duration (s)	Cycles counted	f_{pedal} (Hz)	Cadence (RPM)
rpm-video-1.mp4	17	26	1.53	91.76
rpm-video-2.mp4	21	30	1.43	85.71
rpm-video-3.mp4	25	30	1.20	72.00

Based on the results computed in table 2, it is clear important to observe that we’re only interested in the highest possible speed, and that is why the group participant was request to cycle as fast as possible. Since a single gear bicycle was used in the experiment, one could claim that the number of leg revolutions would be as high as possible and still, the maximum frequency observed was still under 2Hz .

The combined results of both experiments offer a solid base for the intuition that there seems to be, indeed, a gap in the frequency spectrum of the leg movements and the road vibration frequency.

[Insert: Place figure at figures/sampling_rates.pdf]

Figure 3: Measured sampling-rate checks across the acquired recordings.

4 Data Preparation

Processing pipeline. [Insert: Sensor alignment, calibrated dynamic acceleration projected along the gravity direction, gyro magnitude, jerk, operational-state gating and 2 s transition exclusion. Explain that centred filters are offline/noncausal.]

Frequency analysis and windowing. [Insert: State the physics-informed 4–46 Hz Butterworth analysis band, Welch PSD, 2 s windows and 50% overlap; do not claim everything inside the band is road-caused.]

Features and reduction. [Insert: Document the 12 engineered candidates; correlation and mutual-information analysis using TRAIN only; motivate compact sets and note redundancy/physical interpretation.]

5 Modeling

Partitioning. [Insert: Describe frozen multi-participant, complete-recording 78/22 hold-out; 13 independent training split groups, including two original parent sessions. Explain grouped internal validation and scaling within each fold.]

Candidate feature sets. [Insert: Describe full_12, compact_physics_5 and compact_ratio_4; justify their comparison.]

[Insert: Place figure at figures/state_gating.pdf]

Figure 4: Operational-state gating and valid riding segments.

[Insert: Place figure at figures/psd_terrain.pdf]

Figure 5: State-gated PSD comparison between Smooth and Bumpy windows.

Supervised methods. k-NN, Logistic Regression, Decision Tree and Random Forest. [Insert: Record the course-motivated reason for each and the actual hyperparameter search ranges.]

Unsupervised methods. K-Means, Agglomerative Clustering, DBSCAN and Gaussian Mixture Model. [Insert: Contrast cluster assumptions; explain that terrain labels are never inputs to clustering.]

Table 3: Model settings and comparison plan (complete only after running the notebook).

Approach	Method	Selected features / parameters / evidence
Supervised	k-NN	[Insert: Fill from validated model run]
Supervised	Logistic Regression	[Insert: Fill]
Supervised	Decision Tree	[Insert: Fill]
Supervised	Random Forest	[Insert: Fill]
Unsupervised	K-Means	[Insert: Fill]
Unsupervised	Agglomerative	[Insert: Fill]
Unsupervised	DBSCAN	[Insert: Fill]
Unsupervised	GMM	[Insert: Fill]

6 Evaluation

Selection criterion. [Insert: Define positive class Bumpy; justify emphasis on Bumpy recall and false negatives alongside precision, F1 and balanced accuracy. Identify whether ROC-AUC was valid for each validation fold.]

Supervised comparison. [Insert: Training-only grouped validation methodology, feature-set comparison, selected hyperparameters, aggregate and per-recording metrics. Distinguish the exploratory pilot from final comparison.]

Clustering comparison. [Insert: Clusters/noise fraction; ARI and NMI using labels only for post-fit diagnostics; silhouette where defined. Discuss out-of-sample limitations of Agglomerative and DBSCAN.]

Frozen hold-out. [Insert: Only after model and threshold selection: report the independent four-recording test results once; analyze missed bumps, false alarms, participant/group variation and dataset limitations.]

7 Deployment Considerations & Recommendations

External deployment. [Insert: Apply the final frozen preprocessing, feature representation, scaler, estimator and decision rule to the professor’s independent external data. The external dataset is not used for fitting, tuning or testing.]

Limitations and recommendations. [Insert: Small independent session count, only two retained participants, group/window dependence, sampling hardware mismatch, offline/noncausal processing, transfer to new devices/routes, and future independent collections. Include consent/anonymization and data deletion according to the assignment.]

References

- [1] Colin Shearer. “The CRISP-DM Model: The New Blueprint for Data Mining”. In: *Journal of Data Warehousing* 5.4 (2000), pp. 13–22.

[Insert: Place figure at figures/final_confusion_matrix.pdf]

Figure 6: Final held-out confusion matrix; Bumpy treated as the positive class.

[Insert: Place figure at figures/external_timeline.pdf]

Figure 7: Predicted Smooth/Bumpy sections on the independent external ride, plotted against elapsed time or distance.